

The Inception of In-Context Learning: Minimal Training Structure for Contextual Candidate Promotion

Paper 0.8 — Controlled ICL Mechanism Research Line

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Abstract

We ask for the smallest training structure and Transformer in which context promotes a target that is low-ranked without demonstrations to rank one. A leakage-audited D0–D4 ladder excludes the critical $C \rightarrow 3$ mapping and compares correct demonstrations with shuffled, wrong-chain, irrelevant, reversal, and no-context controls. We first classify the computational tasks: the present D4 is finite permutation constraint completion (T3), not rule induction. Under its uniform k -symbol permutation prior, all-but-one demonstrations reduce $H(Y | X) = \log_2 k$ to $H(Y | X, D_{\text{ctx}}) = 0$, so the context supplies $\log_2 k$ bits and the oracle ceiling is one. A 54-cell coarse sweep finds 16 individual positives, but none of five Pareto-small candidates passes all three seeds. Contextual-copy calibration recovers early pair binding and later associated-value transport in one competent checkpoint, yet fresh replications also fail the 95% gate. We then run a learned D5 modular-rule test, not merely a generator audit: four SA-only/SA+FF architectures, three seeds, seen and held-out parameters, and a held-out square family on $\mathbb{Z}_{5,7,11}$. All 2,544 generated train/evaluation episodes are exactly identifiable and leakage-free. No cell passes; correct-context accuracy remains nonselective against no-context, shuffled, and wrong-rule controls. Thus information sufficiency repeatedly exceeds reliable acquisition in these tiny models, and a physical copy mechanism available after acquisition should not be conflated with parametric rule induction.

1 Question

Let Y be a candidate with low context-free rank, $\text{rank}(Y | X) \gg 1$. We seek the smallest training distribution $\mathcal{D}^*$ and smallest model M^* for which $\text{rank}(Y | D_{\text{ctx}}, X) = 1$, while Y remains low-ranked without D_{ctx} . Define

$$G_{\text{ICL}}(Y) = m(Y | D_{\text{ctx}}, X) - m(Y | X), \quad m(Y) = z(Y) - \max_{y \neq Y} z(y).$$

The threshold is crossed when contextual margin becomes positive on held-out relations and the effect disappears under matched shuffled/wrong-context controls.

2 Minimal motivating example

Train local orders $A \rightarrow B, B \rightarrow C$ and $1 \rightarrow 2, 2 \rightarrow 3$, but never train $C \rightarrow 3$ or the complete paired mapping. Test

$$A \ 1, \quad B \ 2, \quad C \ ?$$

and ask whether context aligns the two local chains strongly enough to promote 3.

3 Minimal-dataset ladder

- D0: random/no useful structure.
- D1: isolated local successor chains only.
- D2: many isomorphic independent chains.
- D3: paired alignment or selection from a finite transformation family.
- D4: episode-specific permutation completion preventing global mapping memorization.
- D5: identifiable parametric or structured rules, introduced only after the simpler regimes are separated.

At each stage add exactly one structural ingredient.

3.1 Computational taxonomy and information sufficiency

We distinguish in-weight association (T0), direct copy (T1), associative/induction copy (T2), finite constraint completion (T3), finite task selection (T4), parametric rule induction (T5), compositional rule induction (T6), and procedural or recursive induction (T7). These labels describe the computation licensed by the generator, not what a trained network necessarily implements. Table 1 locates the present ladder. D1–D2 test whether local relational exposure creates contextual alignment pressure; the current paired D3 is a finite alignment/completion test. Proposed D3-B/C variants would instead select among a known finite mapping or transformation set and must separate seen from unseen mappings.

Table 1: Dataset taxonomy. “Determined” means that context plus the declared prior fixes the answer; it does not imply that a model learned the computation.

Regime	Type	Computational task	Target shown	Determined	Iterative
D0–D2	T0	association/local structure	no	no	no
D3-A	T3	partial alignment completion	no	yes	no
D3-B/C	T4	finite task selection	no	yes	no
D4	T3	permutation constraint completion	no	yes	no
D5 unary	T5	parametric modular rule	no	gated	no
D5 composed	T6/T7	compositional/procedural rule	no	gated	possibly

For D4, let π be uniform over the $k!$ permutations and let the query identify one key. Before demonstrations, its value is uniform, hence

$$H(Y | X) = \log_2 k.$$

The prompt shows $k - 1$ key–value pairs. The omitted value—which is not explicitly present in the context—is uniquely fixed by permutation complement, giving

$$H(Y | X, D_{\text{ctx}}) = 0, \quad I(Y; D_{\text{ctx}} | X) = \log_2 k,$$

and Bayes/oracle accuracy one. In the experiments $k = 3$: the mapping space has $3! = 6$ members, each prompt contains two demonstrations, and the context contributes $\log_2 3 \simeq 1.585$ bits. Consequently, a selective D4 positive establishes genuine contextual dependence, but specifically for T3 elimination/completion; it is not evidence of a newly inferred function.

3.2 D5 validity gate

We implemented a CPU-only exact-posterior validator for successor, add- n , multiply-by- a , affine, and square functions on $\mathbb{Z}_k$, for $k \in \{5, 7, 11\}$. Each episode enumerates the declared finite hypothesis family, filters parameters consistent with the demonstrations, computes the induced answer distribution at the query, and rejects the episode unless that answer has zero entropy. It also stores the number of consistent hypotheses and the minimum identifying prefix. Across 360 validation episodes (24 per family/modulus cell), all passed with posterior answer entropy zero; 284 required one demonstration and 76 required two. The latter comprise all 72 affine episodes and four multiply-by- a episodes. This first audit verifies generator admissibility only. The learned experiment below strengthens the prior by enumerating the joint family-and-parameter hypothesis set rather than validating each family in isolation.

4 Dataset and model phase diagrams

Search over number of relation families, examples per family, task/mapping entropy, chain length, symbol reuse, template variation, and position variation. Start with

$$L \in \{1, 2, 3, 4, 6\}, \quad d_{\text{model}} \in \{8, 16, 32, 64\},$$

and one or two heads where feasible. The goal is a behavioral phase boundary, not maximum benchmark performance.

5 Controls

Every positive cell compares no demonstrations, correct demonstrations, shuffled outputs, wrong-chain demonstrations, same-length irrelevant context, order perturbations, random initialization, held-out symbols, and held-out relation families. Audit that the critical direct mapping never occurs in training and that the target remains low-ranked without context.

6 Layerwise mechanism

At every boundary record residual state, SA update, FF update, target logit, rank, margin, top- k candidates, and entropy. Compare context-free and context-conditioned trajectories. The first question is where Y becomes a serious candidate; the second is where it becomes rank one.

7 Mechanistic hypotheses

Test whether attention aligns corresponding relational roles, transports a successor/order state, or directly increases support for the target. Test whether FF already encodes reusable successor transformations, maps an attention-transported relation into the target direction, or simply sharpens a candidate. These are competing hypotheses.

8 Causal experiments

Ablate heads; zero or replace SA/FF updates; patch states from correct, shuffled, wrong, role-matched, and matched-norm controls; block individual demonstration tokens; and remove one local relation from training and retrain. Prefer causal loss of contextual promotion over attention-map interpretation.

9 Before/after acquisition

Find closely matched $\mathcal{D}^-$ and $\mathcal{D}^+$ conditions immediately below and above the ICL threshold. Compare head utility, SA/FF updates, residual target projection, relation-role similarity, cross-layer covariance, and weights. This threshold comparison is central.

10 Minimality

After finding a positive regime, delete one training family, relation type, positional regularity, symbol overlap, layer, head, or width. If the effect survives, simplify again.

11 Interpretation

Do not assume the Transformer is performing regression, Bayesian inference, or implicit gradient descent. Those may be useful behavioral approximations at larger scale. Prefer the smallest measured mechanism consistent with the evidence:

learned local relations + contextual alignment/transport + candidate promotion.

12 D0–D4 behavioral results

The reference L2/W32/H2 model was evaluated with three seeds. D0 never passes. D1 and D2 reach 87.5% correct-context accuracy, but their controls can perform equally well; local order is therefore insufficient. D3 and D4 reach 100% correct-context accuracy, yet only one of three seeds passes the complete context-dependence gate. Across the full 54-cell coarse surface, 16 cells pass individually.

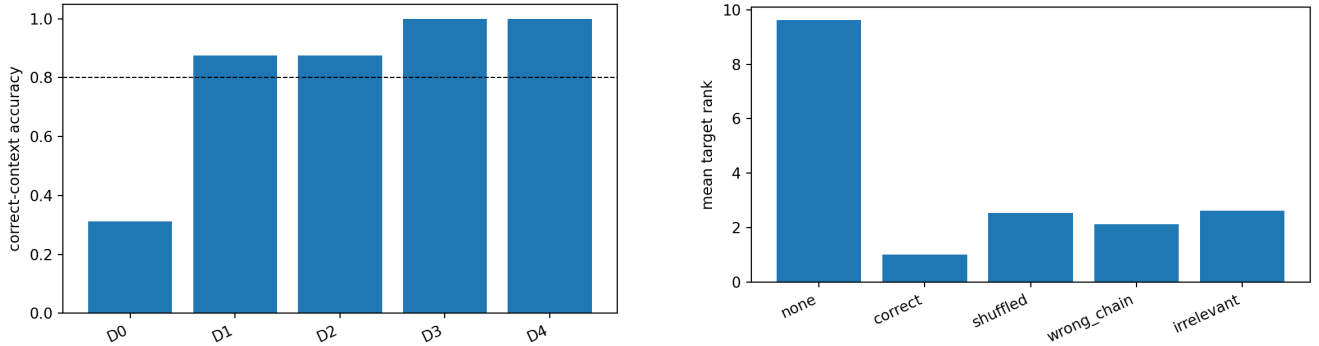


Figure 1: Left: dataset-ladder accuracy at the reference architecture. Right: D4 seed-11 rank promotion and matched controls. High correct-context accuracy is insufficient when a control also succeeds or the context-free target is already favored.

The smallest local positive is L1/W16/H1: 100% correct context, zero context-free accuracy, mean context-free rank 26.5, zero negative-control accuracy, and margin gain 18.37. Replication rejects it: seeds 23 and 37 achieve only 25%. Four further Pareto candidates also fail the three-seed gate. L2/W16/H1 and L6/W8/H1 pass two seeds; all others pass one. The supported result is a sharp, non-monotonic, seed-sensitive acquisition frontier. D4 tracing is consequently blocked rather than selected from a favorable seed.

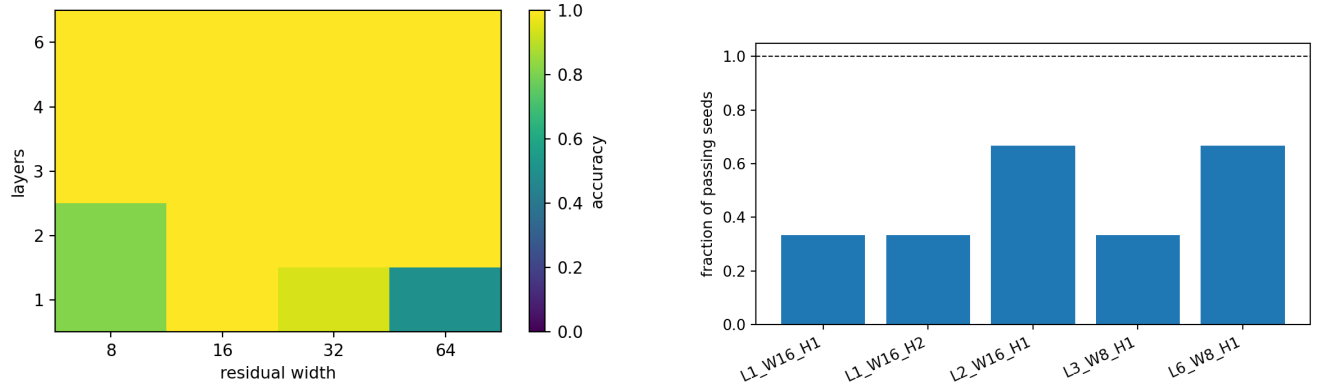


Figure 2: Coarse seed-11 architecture surface (left) and three-seed replication fraction for Pareto-small positives (right). Deeper or wider models are not monotonically more selective.

13 Contextual-copy calibration

The C2 calibration uses balanced episode-specific associations, so a fixed key-to-value map cannot solve the query. One L2/W64/H2 checkpoint passes at 100%, context-free rank 3.0, and 14.6% maximum association-control accuracy. Exact Q/K/V tracing finds layer-0 head-0 previous-token mass 0.441. At layer 1, the two heads place 0.624 and 0.664 attention probability on the associated demonstration value, while repeated-key attention is only 0.039 and 0.068. This is a pair-binding then value-routing circuit, not a claim that attention probability itself is mass flow.

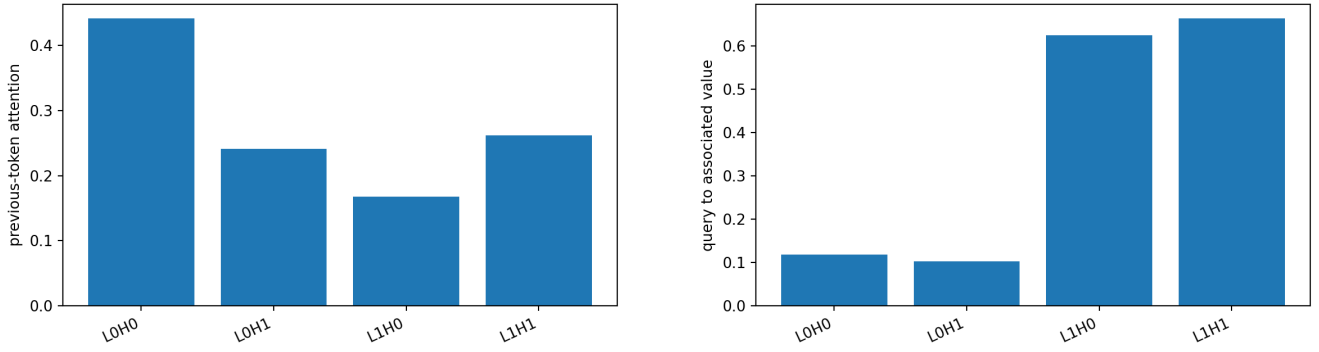


Figure 3: Role-aligned copy calibration: early previous-token binding and later query-to-associated-value routing.

Causal tests agree. Layer-1 demonstration-value ablation lowers top-1 accuracy from 100% to 37.5% and damages margin by 15.24. Patching all layer-1 values from shuffled context lowers accuracy to 20.8% (margin damage 24.89); patching keys retains 77.1%. Layer-0 head-0 ablation lowers accuracy to 54.2%, while both layer-1 heads are useful. Mean margin progresses from -28.33 initially through -21.98 after layer-0 SA and -3.83 after layer-0 FF, then to $+15.54$ after layer-1 SA. Final FF preserves rank one but slightly reduces margin to $+13.65$. Thus later attention supplies the decisive target support.

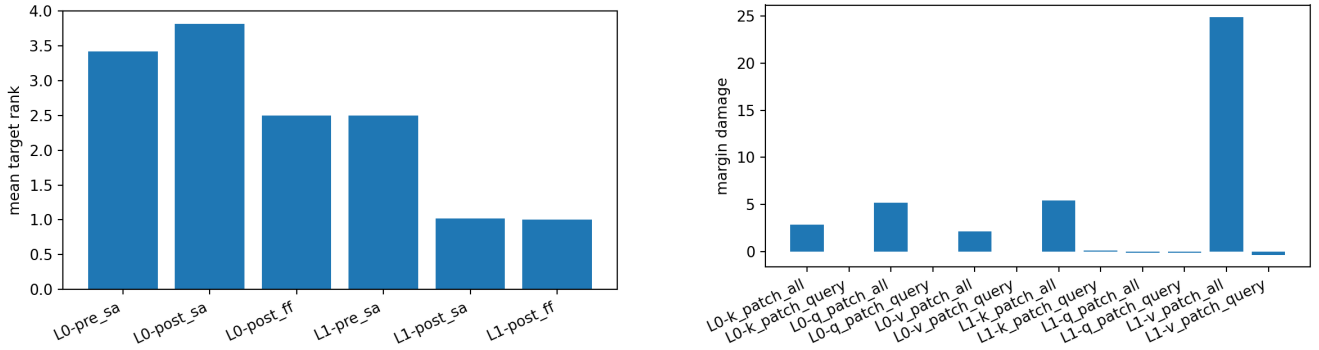


Figure 4: Target trajectory (left) and causal Q/K/V patch damage (right). The target becomes rank one principally in second-layer attention, and shuffled value transport is most destructive.

This is a local instrumentation calibration. Fresh L2/W64/H2 replications at 256 episodes reach 89.6%, 70.8%, and 66.7%, all below the 95% gate; L2/W128/H2 reaches only 41.7%, 54.2%, and 33.3%. Increasing episode diversity to ($E=1024$) makes seeds 11 and 23 competent at both the nominal and doubled update budgets (100% each), but seed 37 reaches only 52.1% and 81.2%, respectively. A targeted fourfold-budget run does not rescue seed 37: accuracy falls to 60.4%. The non-monotonic trajectory rules out the simple explanation that this failure merely needed more updates. It instead supports an initialization- and optimization-sensitive acquisition boundary. Hence the circuit is physically validated where acquired, while reliable three-seed acquisition remains unestablished.

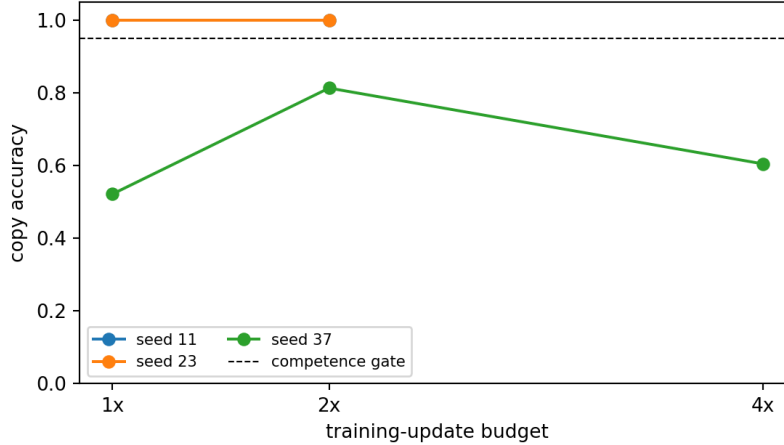


Figure 5: Copy acquisition at ($E=1024$) versus update budget. Seeds 11 and 23 pass at the measured 1x and 2x budgets; seed 37 improves at 2x but regresses at 4x. The isolated 4x point is a targeted failure diagnostic, not a matched three-seed estimate.

14 Learned D5 modular-rule acquisition

The D5 first pass asks whether the same small model can use demonstrations to infer a unary modular rule rather than complete a permutation. Inputs and outputs occupy modulus-specific numeric vocabularies, but no family or parameter name is serialized. Training mixes successor, add- n (including add-two), multiply- a , and affine maps on $\mathbb{Z}_k$ for $k \in \{5, 7, 11\}$; square is withheld as an entire function family. Within add, multiply, and affine, a deterministic 25% parameter split is also withheld. Evaluation therefore separates novel episodes using seen parameters, held-out parameters from seen families, and the held-out square family.

For every episode, the validity gate enumerates the union of all five declared hypothesis families, adds demonstrations until the query answer is unique, and rejects residual answer entropy. The target query pair is never demonstrated. Train and evaluation contain 1,472 and 1,072 episodes respectively, with zero serialized-episode overlap, zero held-out-parameter overlap, and no square episode in training. All 2,544 episodes have Bayes ceiling one and require at most three demonstrations. This validates the information claim while avoiding a family-name lexical shortcut; it does not force the learned model to recover the generating family when several hypotheses predict the same answer.

We train four first-pass architectures—L1/W32/H1 and L2/W64/H4, each SA-only and SA+FF—for 1,200 updates and seeds 11, 23, and 37. Each correct prompt is paired at evaluation with fixed-length no-context, shuffled-output, and wrong-rule controls. A seed passes a split only if correct-context accuracy is at least .95 and exceeds every matched control by at least .20.

No seed in any architecture passes any split. Three-seed mean seen-parameter accuracy ranges from .115 to .130 across SA-only models and from .118 to .119 across SA+FF models, near the .121 mean uniform baseline. Held-out-parameter means range from .088 to .153, again without positive selectivity. Held-out-square accuracy is sometimes numerically higher (.122–.250 by architecture), but the corresponding controls rise with it: mean selectivity ranges from $-.056$ to zero. Even the best individual correct-context accuracies—.151 seen-parameter, .158 held-out-parameter, and .302 held-out-family—fail because a matched control is as good or better. Family-level accuracy likewise remains nonselective; this experiment supplies no learned T5 rule-induction positive.

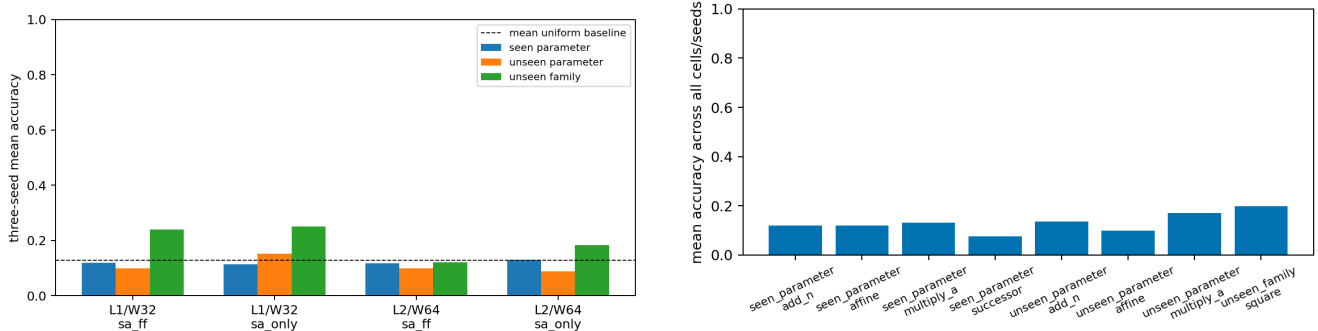


Figure 6: Learned D5 first pass. Left: three-seed mean correct-context accuracy by architecture and evaluation split; the dashed line is the mean uniform baseline. Right: accuracy by family, pooled over the four architectures and three seeds. Neither panel is a competence claim: matched controls eliminate every apparent elevation.

The negative is stronger than generator-only validation but narrower than a capacity result. Final training losses remain 1.31–3.01 across cells, so the mixed curriculum is not reliably acquired even on its training objective; the experiment does not distinguish insufficient optimization from an architectural limitation. The next justified step is a curriculum ladder beginning with one parametric family and explicit within-family generalization, followed by controlled mixtures. Mechanistic tracing remains gated because there is no selectively competent D5 checkpoint.

15 Answers and limitations

Local order alone is not sufficient. Partial/fresh mapping pressure creates local positives, but no tested D4 architecture is seed-stable. D4 is nevertheless information-sufficient: its context determines the held-out value exactly. The D5 experiment now extends this separation to identifiable modular functions: all inputs pass the exact information gate, while none of 12 model/seed cells acquires selective behavior even for seen parameters. This is an acquisition/optimization failure in the tested mixed curriculum, not evidence that T5 is unrepresentable. Where contextual copy is acquired, the smallest measured mechanism is *bind pairs*, *match the query role*, *transport the associated value*, *promote its logit*; second-layer SA is decisive and FF is supportive rather than the final promoter. Increasing copy diversity rescues two seeds, but update-budget scaling is non-monotonic for the remaining seed and does not close the replication gate. We find no need for regression, Bayesian, or implicit-gradient metaphors. We do not claim a stable smallest copy, D4, or D5 model, and trace no mechanism from a selectively chosen D5 seed.

Table 2: Final evidence audit. A behavioral positive is not promoted to a stable mechanism unless its stated replication gate passes.

Component	Evidence	Gate	Supported interpretation
D0–D4 generator	exact audits and controls	pass	valid bounded task ladder
D4 behavior	16/54 seed-local positives	0/5 architectures stable	acquisition frontier only
Copy behavior	competent traced checkpoint	fresh three-seed gate fails	local calibration
Copy mechanism	tracing plus causal patches	pass in that checkpoint	pair binding/value routing
D5 behavior	12 learned cells; 2,544 valid episodes	0 passing seeds	no mixed-curriculum acquisition

16 Reproducibility

Generators and phase runners are under `src/cl/semantic/` and `src/cl/experiments/`. The shared tracer and analysis modules use the `paper08_*` prefix. Configurations are under `configs/paper08/`. The taxonomy audit is configured by `taxonomy_v1.json`; its exact D5 posterior rows and machine-readable D0–D5 classification are under `results/taxonomy/`. Learned D5 uses `d5_acquisition_v1.json`, `paper08_d5.py`, `paper08_d5_acquisition.py`, and `paper08_d5_analyze.py`; authoritative CSV/JSON/PNG artifacts are isolated under `results/d5_acquisition_v1/`. Its manifest records 12/12 completed CPU cells and all leakage gates. Tracked artifacts also include raw phase/control rows, five three-seed D4 replications, copy replication and exact trace ta-

bles, the (E=1024) training-budget diagnostic, and compressed Q/K/V matrices. Dense checkpoints remain locally reproducible and git-ignored.

17 Relation to Papers 0.5–0.7

Paper 0.5 studies predictive refinement once evidence is accessible. Paper 0.6 studies learned semantic/compositional relations. Paper 0.8 asks how context combines learned relations to create a new prediction without weight updates. Paper 0.7 now supplies validated formal-task infrastructure, while its learned formal-model frontier remains pending; no Paper 0.7 result is used as evidence here.

18 Scaling policy

Do not scale merely to improve performance. Scale only if the minimal model cannot express the capability, a mechanistic hypothesis requires additional components, or external validity becomes the explicit question.

19 Contribution boundary

The present contribution is not a stable minimal ICL or rule-induction model. It is a disciplined boundary: an information-valid constraint-completion task whose observed positives remain seed-sensitive, a contextual-copy checkpoint in which causal interventions locate binding and value routing, and an information-valid D5 mixed curriculum whose learned cells remain at nonselective baseline. A future positive must pass the matched controls across seeds before its signed target-logit support is traced. The negative D5 result motivates simplifying the curriculum, not enlarging the theoretical claim.